Reranker diagnosis and tested measures

Analysis date: 14 October 2026. Latency values measure the reranking stage only.

Findings

The main cost is repeated local transformer inference. The current reranker is a cross-encoder: it reads the question together with every candidate document and performs a separate relevance calculation for each pair. The BGE model card describes this query/passage scoring behavior. No network request or graph database command occurs inside this model stage.

The cached model has 278,044,417 parameter elements, 12 layers, 768 hidden dimensions and float32 weights. Its file occupies 1,112,459,588 bytes. The tokenizer permits 512 tokens per question/document pair. The original export runs attention using separate matrix operations. Operator profiling placed all observed heavy MatMul, FusedMatMul and Gemm operations on CUDA. The CPU operators were small shape operations such as Gather, Unsqueeze and Reshape.

The application sends complete conversation documents to the tokenizer and scores the whole candidate pool. DMR scores 100 candidates per query, LoCoMo 19–32, and LongMemEval 38–56. With batch size 16, DMR requires seven model calls. The final result limit of ten is applied after scoring; changing only top_k does not reduce the number of documents sent to the model.

Test scope

The experiments retain all 597 questions from the previous 20% cohorts and their exact candidate pools. Each of three variants scores every question: 1,791 measured reranking requests. The original variant reproduces every previously returned ID, ordering and predicted answer. All 17 quality metrics were recomputed. DMR and LongMemEval each have 100 scored questions. LoCoMo has 397 questions, 395 retrieval-scored examples and 328 answer-scored examples; missing/empty references are excluded from the corresponding metric denominators.

The application source is pinned to the revision used in the preceding latency report. The current reranker file has no working-tree changes. Newer lifecycle/shared-memory changes are outside this experiment. Candidate pools come from private copies of the original embedded vector stores. The verified benchmark graphs have no entities or edges, so graph lookup is represented as empty when constructing those pools. Original baseline output equality verifies that this reconstruction gives the same reranked results on every question.

BAAI/bge-reranker-base runs locally with ONNX Runtime 1.23.2 on the RTX 4050 Laptop GPU. Model loading, conversion, embedding, vector search, Redis checks, graph operations and result generation/scoring are excluded from reranking latency. All variants use the same deterministic shuffled question order, two warm-up requests, and one complete measured pass. The FP32 variants run first; the session is released before loading FP16 to avoid holding two large sessions. This fixed variant order does not control thermal/power conditions. GPU snapshots are retained in full_telemetry.jsonl; they are instantaneous samples, not kernel timings or peak memory.

Time inside the original reranker

Dataset	Total ms	Tokenize	Build inputs	ONNX run	Other
dmr	1717.07	34.83	7.78	1668.69	5.77
locomo	394.05	17.57	1.89	372.53	2.05
longmemeval	940.67	113.19	3.90	815.01	8.57

ONNX run includes model execution, input/output transfers and synchronization. Tokenization runs on the CPU and processes the supplied text before limiting the encoded input. Other includes constructing pairs, collecting scores, sorting and timer overhead. The operator profile records provider placement and host node durations; it does not use CUPTI and must not be interpreted as a separate GPU kernel time breakdown.

Dataset	Mean model batches	Mean padded tokens	Mean active tokens	Padding share
dmr	7.00	50411	36107	28.4%
locomo	2.00	11287	11215	0.6%
longmemeval	3.47	24676	24090	2.4%

Full-cohort measures tested

Original uses FP32, batch 16, 512-token inputs and the full candidate pool. The candidate limit scores the first 20 candidates in the original vector retrieval order; remaining candidates are omitted. Fused FP16 keeps every candidate and the same 512-token limit, but uses an offline optimized mixed-precision model.

Dataset	Original ms	20 candidates ms	Fused FP16 ms	FP16 reduction
dmr	1717.07	333.96	564.46	67.1%
locomo	394.05	344.86	137.30	65.2%
longmemeval	940.67	401.56	367.93	60.9%

The optimizer fused 12 Attention, 12 BiasGelu and 24 SkipLayerNormalization operators. The converted file occupies 556,137,378 bytes. This experiment combines operator fusion and reduced precision, so it does not isolate their individual speed contributions. ONNX Runtime's transformer optimizer supports offline fusions and mixed precision when normal session optimization leaves some patterns unfused. The initial generic FP16 conversion did not pass model type validation; only the ONNX Runtime transformer conversion was successfully benchmarked.

Quality changes

Values below are fractions, with each variant compared against the original model. The full quality_metrics.csv includes all 17 metrics and their numerical differences. A change in ordering can occur even when aggregated quality values remain unchanged.

Dataset	Variant	Recall@10	MRR	nDCG@10	Token F1	Exact top ten
dmr	baseline	0.1400	0.0316	0.0568	0.0338	100/100
dmr	cap20	0.1100	0.0259	0.0456	0.0437	0/100
dmr	fp16_optimized	0.1400	0.0316	0.0568	0.0338	81/100
locomo	baseline	0.8962	0.7174	0.7456	0.0203	397/397
locomo	cap20	0.8913	0.7124	0.7402	0.0201	311/397
locomo	fp16_optimized	0.9046	0.7182	0.7481	0.0203	324/397
longmemeval	baseline	0.9455	0.7562	0.7982	0.0253	100/100
longmemeval	cap20	0.9630	0.7666	0.8102	0.0256	9/100

Dataset	Variant	Recall@10	MRR	nDCG@10	Token F1	Exact top ten
longmemeval	fp16_optimized	0.9455	0.7570	0.7990	0.0260	48/100

cap20: 40 of 51 dataset/metric values changed by more than 1e-12. The exact deltas are exported, including changes in both directions.

Dataset	Metric	Difference
dmr	average_precision@10	-0.005643
dmr	mrr	-0.005643
dmr	ndcg@10	-0.011262
dmr	ndcg@3	+0.005000
dmr	ndcg@5	-0.011789
dmr	precision@10	-0.003000
dmr	precision@3	+0.003333
dmr	precision@5	-0.006000
dmr	recall@10	-0.030000
dmr	recall@3	+0.010000
dmr	recall@5	-0.030000
dmr	substring_match	+0.010000
dmr	token_f1	+0.009915
locomo	average_precision@10	-0.005819
locomo	mrr	-0.005024
locomo	ndcg@1	-0.005063
locomo	ndcg@10	-0.005386
locomo	ndcg@3	-0.007654
locomo	ndcg@5	-0.007806
locomo	precision@1	-0.005063
locomo	precision@10	-0.000253
locomo	precision@3	-0.004219
locomo	precision@5	-0.002532
locomo	recall@1	-0.005063
locomo	recall@10	-0.004852
locomo	recall@3	-0.010759
locomo	recall@5	-0.012025
locomo	token_f1	-0.000197

Dataset	Metric	Difference
longmemeval	average_precision@10	+0.009560
longmemeval	mrr	+0.010377
longmemeval	ndcg@10	+0.011980
longmemeval	ndcg@3	+0.012961
longmemeval	ndcg@5	+0.024361
longmemeval	precision@10	+0.001000
longmemeval	precision@3	+0.010000
longmemeval	precision@5	+0.012000
longmemeval	recall@10	+0.017500
longmemeval	recall@3	+0.023333
longmemeval	recall@5	+0.053333
longmemeval	token_f1	+0.000322

fp16_optimized: 13 of 51 dataset/metric values changed by more than 1e-12. The exact deltas are exported, including changes in both directions.

Dataset	Metric	Difference
locomo	average_precision@10	+0.000981
locomo	mrr	+0.000759
locomo	ndcg@10	+0.002522
locomo	ndcg@3	-0.000594
locomo	ndcg@5	-0.000016
locomo	precision@10	+0.001013
locomo	precision@3	-0.000844
locomo	recall@10	+0.008439
locomo	recall@3	-0.000633
longmemeval	average_precision@10	+0.000746
longmemeval	mrr	+0.000746
longmemeval	ndcg@10	+0.000766
longmemeval	token_f1	+0.000667

Smaller pilot experiments

Eight deterministic questions per dataset were also used to test batches of 8 and 32, a 40-candidate limit and a 256-token limit. These pilot timings are descriptive, not confidence intervals. Variants were executed in separate blocks, so small timing differences can reflect changing runtime conditions. Shorter input limits and reduced

candidate pools change the information available to the reranker.

Variant	DMR ms	LoCoMo ms	LongMemEval ms
baseline	1690.44	357.86	947.97
batch8	1567.52	348.53	935.18
batch32	1728.91	367.32	977.17
cap20	344.36	391.75	405.32
cap40	681.85	364.56	768.42
length256	712.83	155.46	441.19
fp16_optimized	554.70	127.24	363.09

Measures in priority order

1. Use a validated fused FP16 model on CUDA as the first performance candidate. It keeps the same model family, candidate pool and input limit. Numerical changes and observed quality deltas still need an explicit acceptance criterion before changing the default. Cache the converted model once, identify it by source-model/runtime hashes, verify finite scores, and keep the original FP32 model available for CPU execution. The prototype already produces and measures the optimized artifact; production loading is unchanged.

2. Add a separate configurable number of candidates to rerank before inference. A final result limit alone does not reduce model work. The measured 20-candidate variant shows the latency/quality tradeoff; apply a cap only where the full metric/category results meet the intended quality requirement. Graph-derived candidates would also need an explicit inclusion policy; these observation-only benchmarks do not test that case.

3. Reduce repeated text preparation by storing relevant passages or caching document tokens with model/tokenizer and document-version keys. This matters especially for LongMemEval after inference is faster. Simply cutting a character prefix is not equivalent to the current tokenizer and could remove relevant evidence. Chunk selection and token caching were not measured in this run.

4. Make reranking optional for requests whose latency requirement is more important than its measured ranking benefit. An adaptive rule based on query/candidate confidence needs calibration and quality evaluation. Cache scores for repeated identical query/document pairs where versions and model identity match; this helps repeats rather than new prompts.

5. Investigate a smaller cross-encoder if a stricter response-time target remains unmet. This changes the model and requires evaluation on the actual memory tasks. No smaller model speed or quality result is claimed here. Batch tuning can be revisited after model optimization; the current pilot does not support increasing batch size as the main fix.

6. For simultaneous requests, move expensive model execution out of the graph lease only after defining consistent snapshot/revision behavior. The previous latency report measured substantial mutex waiting because the existing lock covers the entire request. This changes concurrency semantics and was not applied or tested here.

Reproduction

Install the isolated inspection dependencies into data/reranker_diagnosis/runtime. Run scripts/diagnose_reranker.py with --phase prepare, then --phase optimize, then --phase full --variants baseline,cap20,fp16_optimized. Run scripts/build_reranker_report.py to recompute quality and generate this document. The retained inputs, original vector stores, pinned source checkout and model cache are required. Large converted models and candidate texts remain under ignored data; report files contain measured summaries and

per-query records. No live graph, original corpus or application setting was changed.

References

BGE model behavior, ONNX Runtime transformer optimization, float16 and mixed precision, profiling limitations.

Measured time comparisons

